

DATA SCIENCE TOOLBOX: PYTHON PROGRAMMING

PROJECT REPORT

(Project Semester January-April 2025)

Customer Segmentation Retail

Submitted by

M Shiva Kumar

Registration No.: **12312331**

Programme: **B.Tech. (Computer Science and Engineering)**

Section: **K23KM**

Course Code: **INT 375**

Under the Guidance of

Anchal Kaundal

(29612)

Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that **M Shiva Kumar** bearing Registration no. **12312331** has completed **INT 375** project titled, “**Customer Segmentation Retail**” under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Anchal Kaundal (29612)

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: **April 12, 2025**

DECLARATION

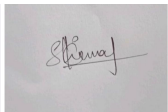
I, **M Shiva Kumar**, student of **Computer Science and Engineering** under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: **April 12, 2025**

Registration No.: **12312331**

Name: **M Shiva Kumar**

Signature

A small rectangular box containing a handwritten signature in black ink. The signature appears to be 'M Shiva Kumar' written in a cursive style.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Lovely Professional University** for providing the academic resources, infrastructure, and an environment that nurtures research and innovation. This project would not have been possible without the institution's unwavering support.

I would also like to acknowledge the use of the **Online Retail Dataset** from the **UCI Machine Learning Repository**, which served as the foundational dataset for this project. Access to such high-quality open data was instrumental in applying and testing various customer segmentation techniques effectively.

Furthermore, I extend my heartfelt appreciation to my **classmates and friends** for their constructive feedback and collaboration, and to my **family** for their endless patience, motivation, and moral support throughout this journey.

This report is the result of consistent effort, learning, and the collective contributions of everyone who guided and supported me.

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INTRODUCTION

In today's data-driven landscape, understanding customer behavior is essential for creating effective business strategies. This project, titled "**Customer Segmentation Retail**," explores how data science can help online retailers identify meaningful customer groups based on purchasing behavior.

Using the **Online Retail dataset from the UCI Machine Learning Repository**, the study analyzes customer transactions through the lens of **Recency, Frequency, and Monetary (RFM)** values. These core metrics are used to evaluate how recently and frequently a customer has purchased, and how much they have spent—offering a powerful foundation for behavioral segmentation.

To deepen these insights, the project applies **K-Means clustering** to group customers into distinct segments. This unsupervised learning technique allows for the discovery of hidden patterns in customer behavior, enabling businesses to:

- Identify high-value customers
- Recognize infrequent or at-risk shoppers
- Tailor marketing campaigns based on data-driven personas
- Enhance customer retention strategies

The project leverages Python libraries including **Pandas**, **NumPy**, **Matplotlib**, **Seaborn**, and **Scikit-learn** to clean, analyze, visualize, and model the data. Through this approach, raw transaction data is transformed into **actionable insights** that support personalized customer engagement and smarter decision-making.

Ultimately, this project demonstrates how customer segmentation can be a cornerstone of modern business intelligence—translating behavioral data into strategies that foster loyalty, growth, and meaningful customer relationships.

Source of Dataset

The dataset used in this project is sourced from a publicly available repository and is widely recognized in the field of customer analytics. It has been made accessible through the **UCI Machine Learning Repository** as part of their initiative to support data-driven research and learning in retail and marketing analytics.

- **Dataset Title:** Online Retail
- **Source:** [UCI Machine Learning Repository – Online Retail Dataset](#)
- **Format:** Excel (.xlsx)
- **Size:** Over 540,000 records, with 8 key fields
- **Time Frame:** December 1, 2010 to December 9, 2011
- **Region Covered:** United Kingdom and select international transactions

Key Features:

- **Transactional Details:** Invoice numbers, product IDs, descriptions, and quantities
- **Customer Information:** Unique customer IDs and associated countries
- **Financial Data:** Unit price and total monetary value of purchases
- **Temporal Data:** Invoice dates for analyzing recency and frequency patterns

Reason for Selection:

- A **credible and curated source** known for real-world datasets
- Enables **RFM-based segmentation** for targeted marketing strategies
- Contains rich and clean transaction data ideal for **unsupervised learning** (e.g., clustering)
- Reflects retail behavior trends that can simulate modern e-commerce environments

Dataset Preprocessing

To ensure the dataset was suitable for customer segmentation, several preprocessing steps were applied using Python's **Pandas**, **NumPy**, and related libraries. These steps addressed data quality issues and prepared the data for **RFM analysis and clustering**.

Key Preprocessing Steps

1. Data Ingestion

- The dataset was loaded from an Excel file using `pd.read_excel()`, resulting in a DataFrame with **541,909 rows and 8 columns**.
- A copy of the raw data (`raw_df`) was preserved to maintain the original state for validation and reproducibility.

2. Duplicate and Null Handling

- **Missing CustomerID values** were removed using `dropna(subset=['CustomerID'])`, which eliminated **135,080 rows** and retained **406,829 records**. This step was crucial, as CustomerID is the primary key for segmentation.
- **Duplicate records** were not removed explicitly, as no exact duplicates were found during exploration.

3. Data Type Conversion

- CustomerID was converted from float to integer using `.astype(int)` to maintain consistency in downstream processing.
- InvoiceDate was already in `datetime64` format, allowing temporal operations like recency calculations.

4. Feature Engineering

- **TotalPrice**: A new column calculated as `Quantity * UnitPrice` to represent the transaction-level revenue.
- **IsCanceled**: A Boolean feature created using `InvoiceNo.str.startswith('C')` to flag canceled or returned transactions for further analysis.
- **Recency**: Calculated as the number of days since the customer's last purchase from the latest transaction date in the dataset.
- **Frequency**: Determined as the count of unique invoices per customer using `nunique()`.

- **Monetary**: Computed as the total value of all transactions per customer using sum() of TotalPrice.

5. RFM Table Creation

- The dataset was grouped by CustomerID to construct an **RFM table**, including:
 - **Recency**: Days since the last transaction
 - **Frequency**: Number of transactions
 - **Monetary**: Total revenue per customer
- Additional metrics like **total orders** and **overall spending** were merged into the final customer profile table.

6. Data Filtering

- **Canceled transactions** were flagged but not discarded, allowing flexibility for comparative analysis between active and canceled orders if needed.

Analysis on Dataset

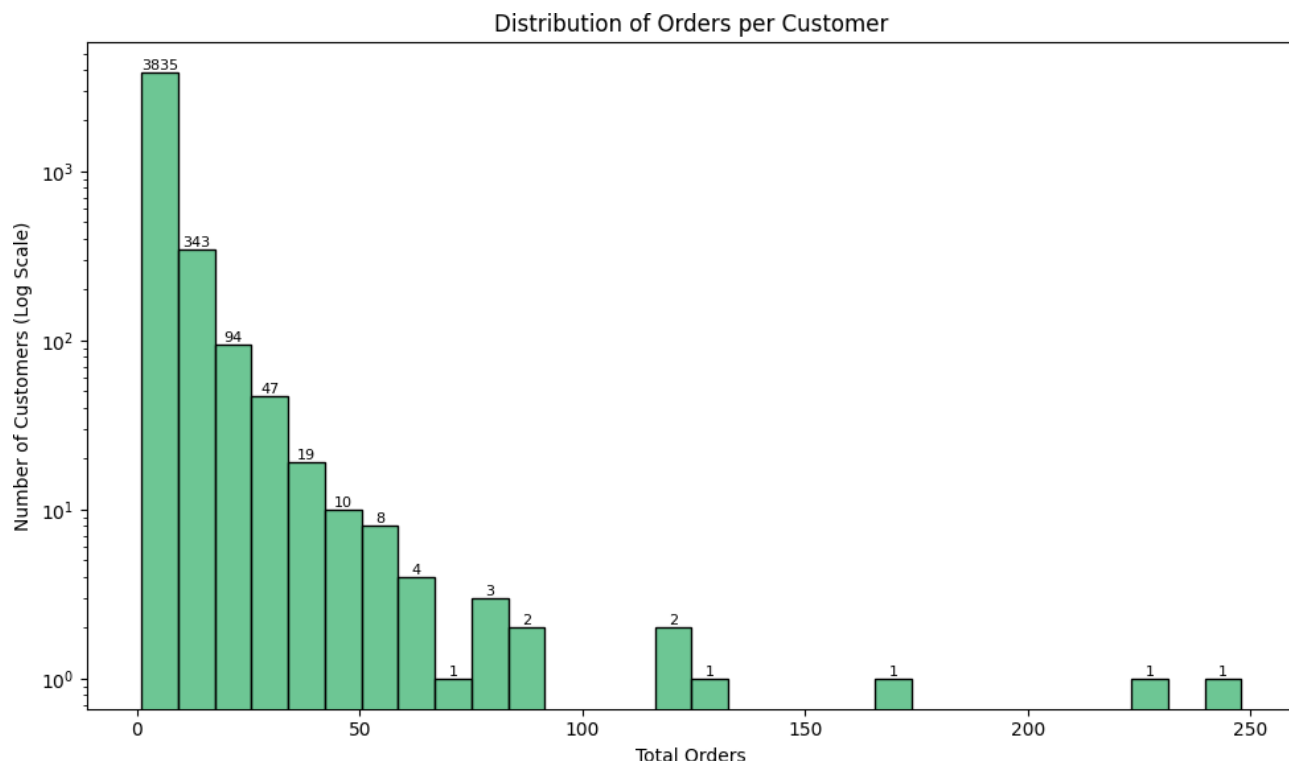
Objective : Calculate Total Spending and Order Frequency per Customer

General Description:

This objective calculates each customer's total spending and order frequency to identify how engaged and valuable each customer is. These two metrics are fundamental to understanding the customer base for segmentation.

Specific Requirements:

The data was grouped by customer ID to calculate two key metrics: total spending, computed as the sum of product quantity multiplied by unit price; and total orders, determined by counting unique invoice numbers per customer. Canceled transactions were excluded to ensure spending reflects only completed purchases.



Analysis Results:

The dataset includes 4,372 customers. Most customers placed between 1 to 5 orders. Spending behavior also varied, with a majority spending less than 1,000 in total. The top 10 customers by total spending were identified, and their contribution to overall revenue is significantly higher compared to the general customer base.

Visualization: Displays a histogram of total orders per number of customer (log scale)

Objective : Top 10 Customers by Total Spending

General Description:

This objective focuses on identifying the top-spending customers based on their total purchase amount. Analyzing total spending helps recognize high-value customers who contribute significantly to revenue and may require personalized attention or rewards.

Specific Requirements:

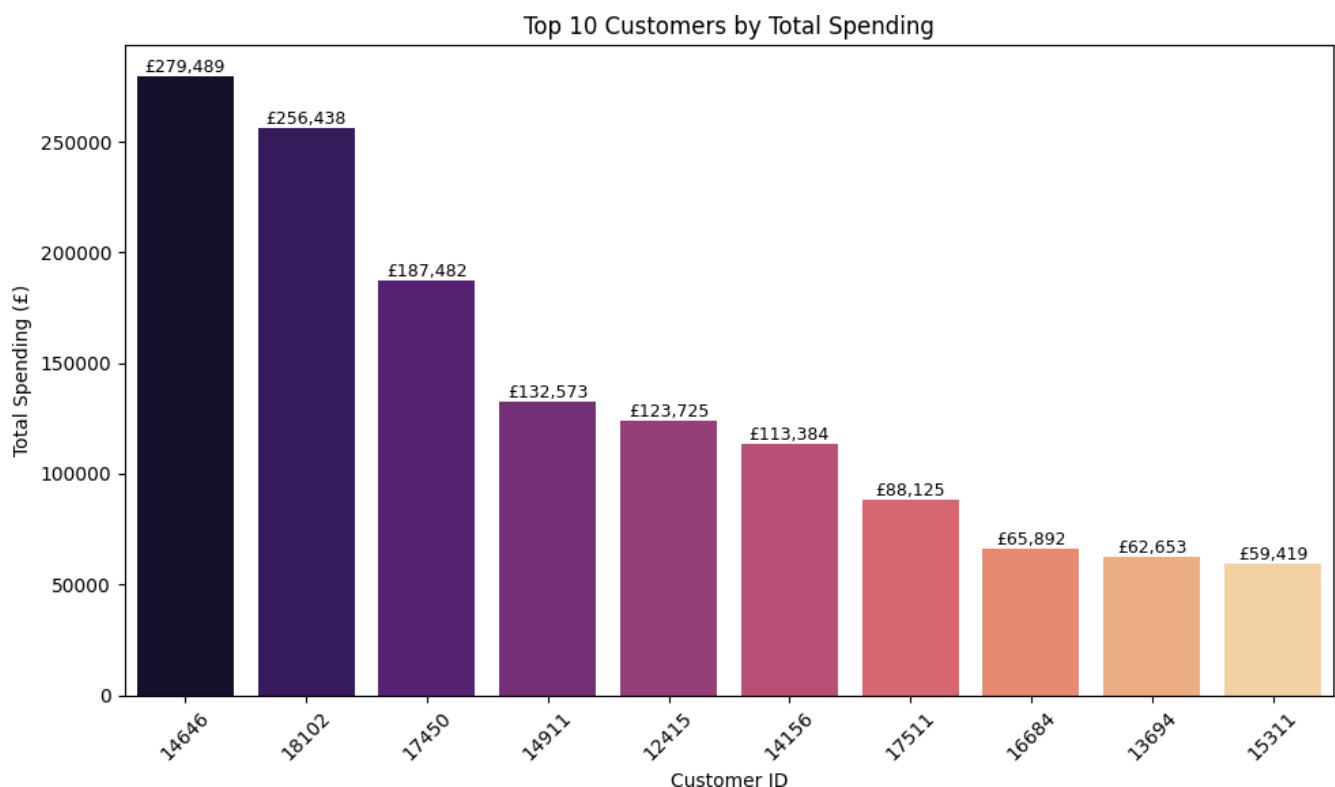
The total spending per customer was calculated by grouping data by customer ID and summing the monetary value of all their valid transactions. Canceled transactions were excluded to ensure accurate values. The top 10 customers were then selected based on the highest total spending.

Analysis Results:

Out of 4,372 customers, the top 10 were identified based on their overall contribution to sales. These customers displayed significantly higher spending compared to the rest, making them valuable for retention-focused marketing strategies. Their individual spending patterns help shape premium segment profiles.

Visualization:

Figure shows a bar plot of the top 10 customers based on total spending. Each bar represents a customer ID and the corresponding amount spent, helping visualize their contribution relative to one another.



Objective : Distribution of Basket Size per Order

General Description:

This objective examines how many items are typically included in each order, also known as the basket size. Understanding this helps in identifying purchasing behavior—whether customers tend to buy in bulk or make smaller, more frequent purchases.

Specific Requirements:

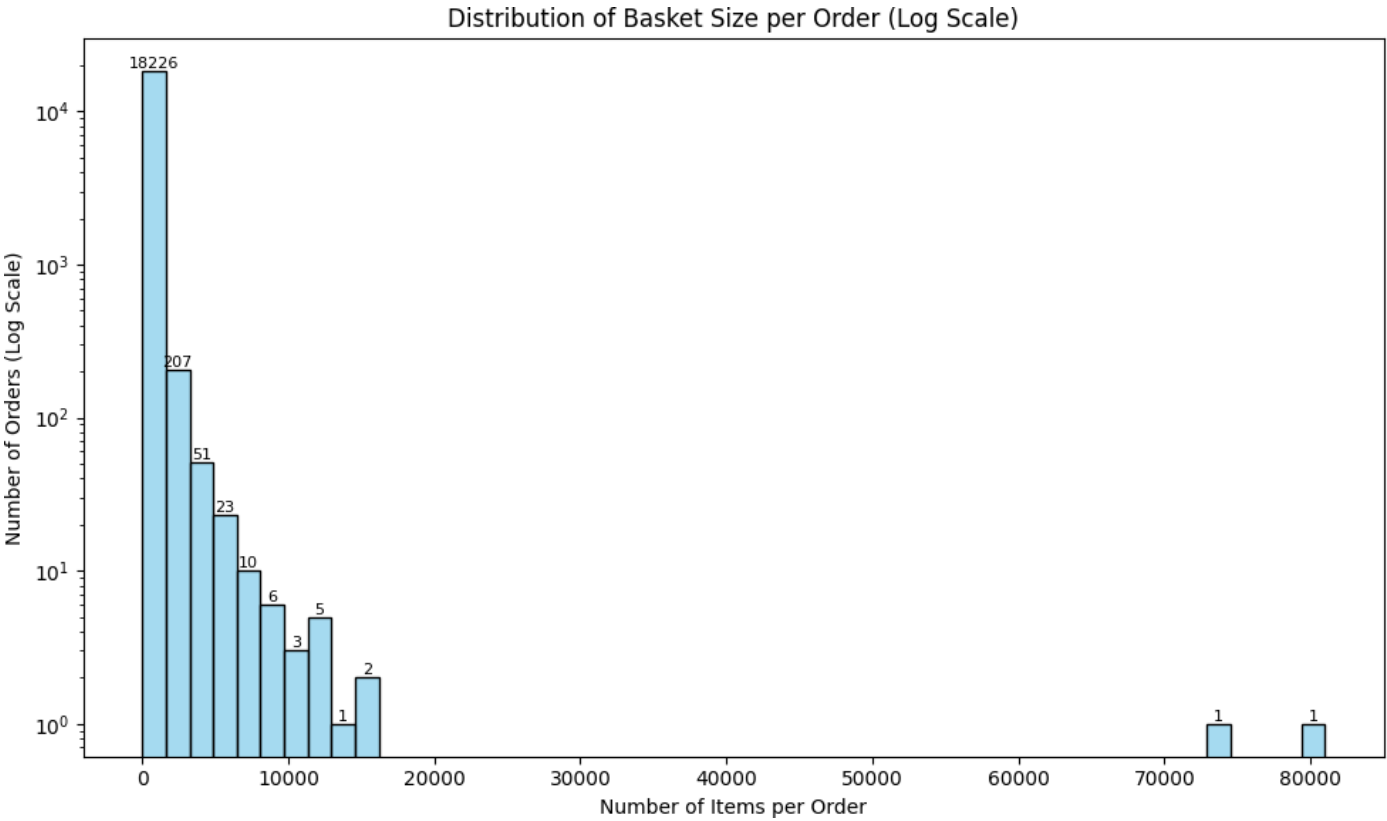
The basket size was calculated by grouping transactions by invoice number and summing the quantity of items in each order. This metric captures the volume of products bought per transaction. Canceled transactions were excluded to ensure that only valid orders were considered.

Analysis Results:

The basket sizes across orders varied widely. Most customers tend to purchase a small number of items per order, while a few place significantly larger orders. This distribution reveals that smaller baskets dominate but there are occasional large-volume purchases.

Visualization:

Figure 1 presents a histogram showing the distribution of basket sizes across all orders. The x-axis represents the number of items per order, and the y-axis shows the count of such orders, plotted on a log scale for clarity.



Objective : Monthly Revenue Analysis

General Description:

This objective examines revenue trends over time by calculating total monthly revenue. It provides insights into business seasonality, peak sales periods, and potential shifts in customer purchasing patterns across different months.

Specific Requirements:

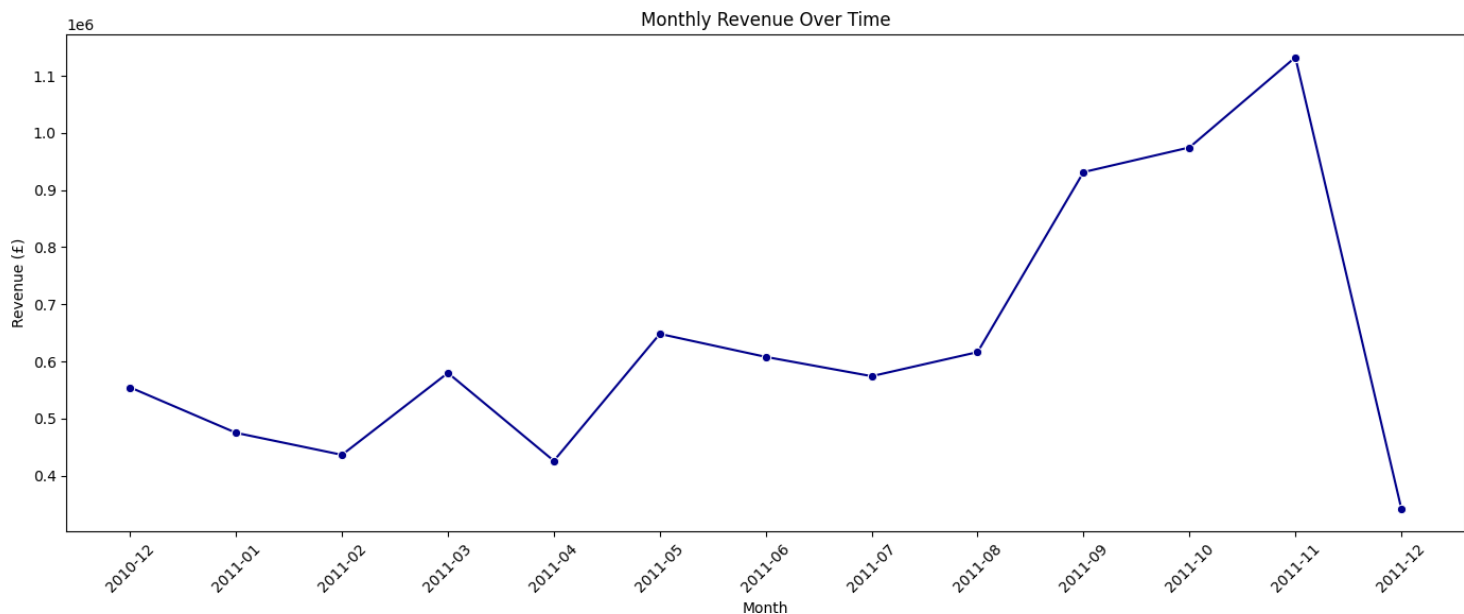
The InvoiceDate was used to extract the month and year of each transaction. Transactions were grouped by these monthly periods, and revenue was calculated by summing the TotalPrice of all transactions per month. Canceled transactions were excluded to ensure accurate revenue representation.

Analysis Results:

Monthly revenue patterns showed fluctuations across the observed period. Certain months experienced noticeable peaks in revenue, indicating successful promotional periods or high-demand seasons. Months with lower revenue could signal off-season performance or operational challenges.

Visualization:

Figure presents a time series line plot of monthly revenue. The x-axis represents the transaction months in chronological order, and the y-axis represents total revenue for each month. The plot highlights seasonal spikes and dips, providing a clear view of revenue trends over time.



Objective : Top 10 Countries by Revenue

General Description:

This objective identifies the top 10 countries contributing the most to total revenue. Understanding geographic distribution helps businesses prioritize market strategies and allocate resources effectively.

Specific Requirements:

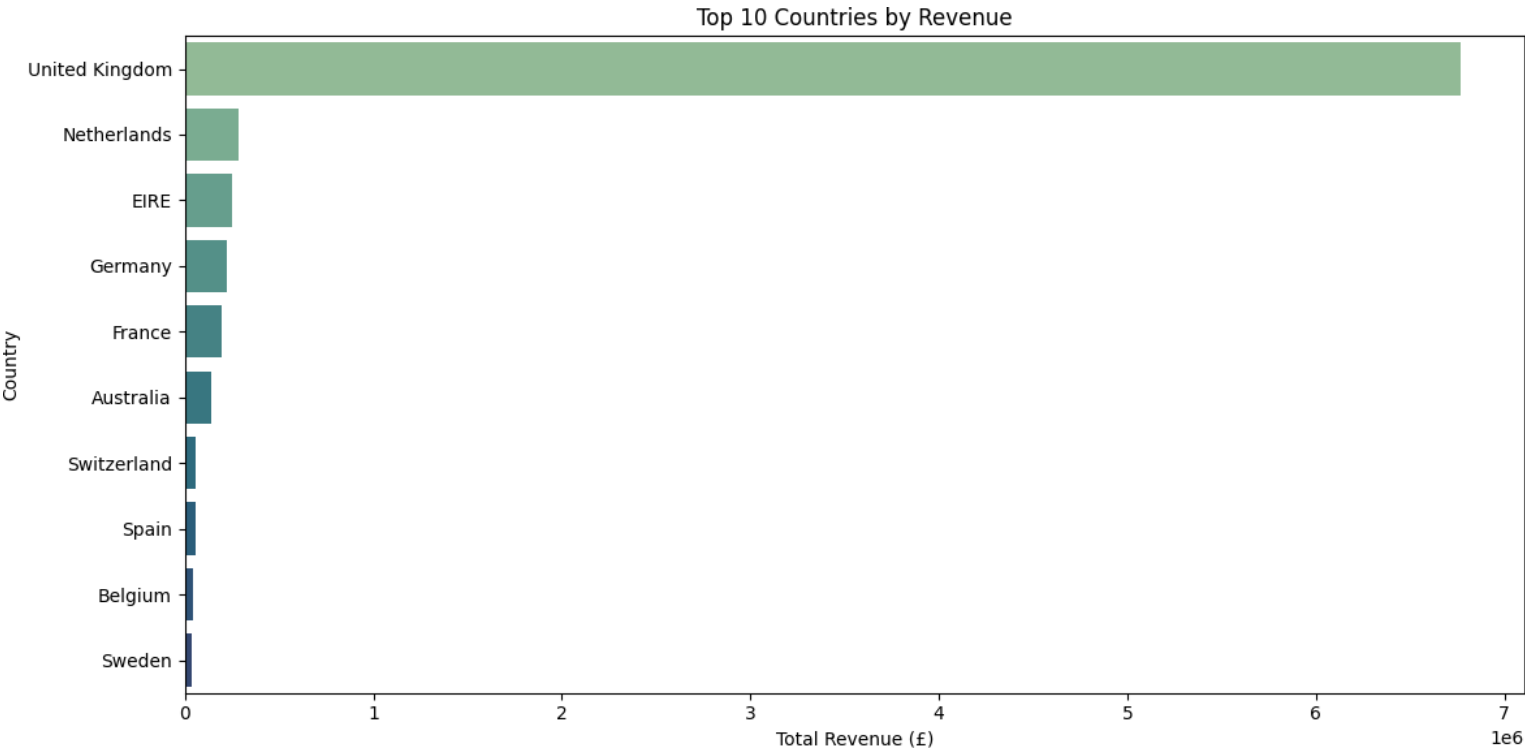
The data was grouped by Country, and revenue was calculated by summing the TotalPrice of all valid (non-canceled) transactions per country. The countries were then ranked in descending order based on total revenue, and the top 10 were selected for visualization.

Analysis Results:

The United Kingdom emerged as the highest contributor to revenue by a significant margin, followed by countries like the Netherlands, Ireland, and Germany. These top 10 countries represent key markets that drive the majority of sales activity.

Visualization:

Figure 1 is a bar plot showing the top 10 countries by total revenue. The x-axis lists the country names, and the y-axis shows their corresponding total revenue. The plot highlights the dominance of UK-based transactions and showcases other high-performing international markets.



Objective : Average Spending per Order by Country

General Description:

This objective focuses on understanding the average monetary value per order in each country. It helps identify markets with higher purchasing power and potential for premium offerings.

Specific Requirements:

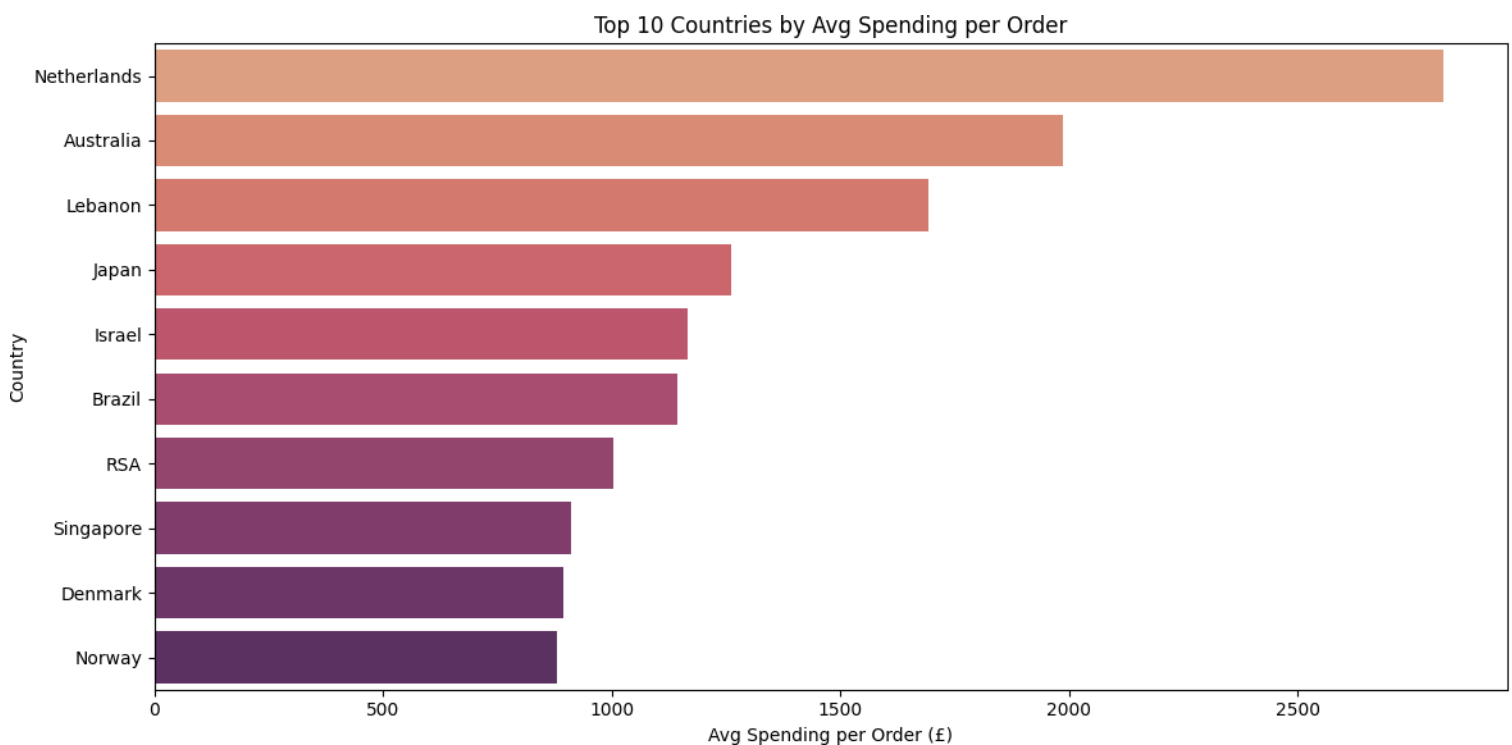
The dataset was grouped by Country, and for each group, total revenue (TotalPrice) was divided by the number of unique orders (InvoiceNo) to compute the average order value. Canceled orders were excluded to maintain data accuracy.

Analysis Results:

Countries such as the Netherlands, Australia, and Sweden ranked high in terms of average order value. Although the United Kingdom had the highest total revenue, several smaller countries had a higher average spend per order, indicating valuable niche markets.

Visualization:

Figure 1 displays a bar plot of the top 10 countries by average order value. The x-axis shows the country names, and the y-axis represents the average spending per order. The chart reveals insights into customer purchasing behavior and market-specific pricing opportunities.



Objective : RFM Analysis and Scoring

General Description:

RFM analysis segments customers based on their Recency (days since last purchase), Frequency (total number

of purchases), and Monetary (total spending). This objective applies RFM scoring to classify customers into behavioral segments that can inform marketing strategies.

Specific Requirements:

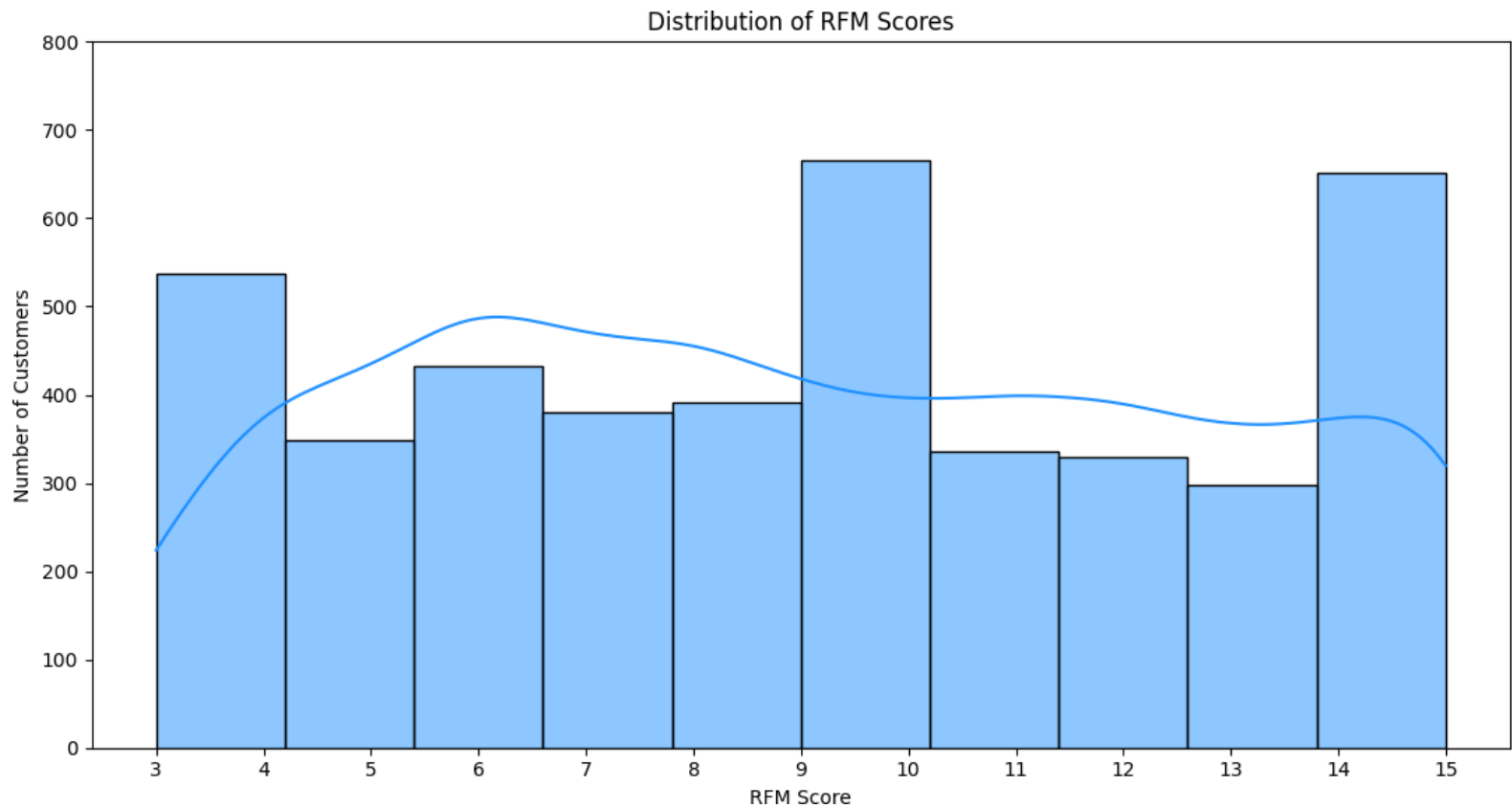
Recency was calculated by subtracting the latest purchase date from a fixed reference date (one day after the most recent transaction in the dataset). Frequency was measured as the number of unique invoices per customer. Monetary value was computed as the sum of all purchases made by the customer. Each RFM metric was assigned a score from 1 to 5 using quintiles, where higher scores represent more desirable customer behavior (e.g., lower Recency, higher Frequency, and higher Monetary).

Analysis Results:

The scoring created standardized RFM profiles, with 5-5-5 representing the best customers: recent, frequent, and high-spending. This system allowed for easy ranking and filtering of customers into groups such as loyal customers, at-risk customers, and high spenders. The RFM scores served as the basis for later clustering steps and strategic targeting.

Visualization:

Figure: RFM Score Distribution per no of Customers
Histogram of total RFM scores ranging from 3 to 15, showing the number of customers in each score bracket.



Objective : Visualizing Customer Segment Counts

General Description:

After clustering customers using their RFM scores, this objective aims to understand how customers are distributed across the identified segments. Segment size helps prioritize marketing focus and customer engagement strategies.

Specific Requirements:

Customers were grouped into four clusters using K-Means based on their RFM scores. A count plot was used to show how many customers fall into each segment.

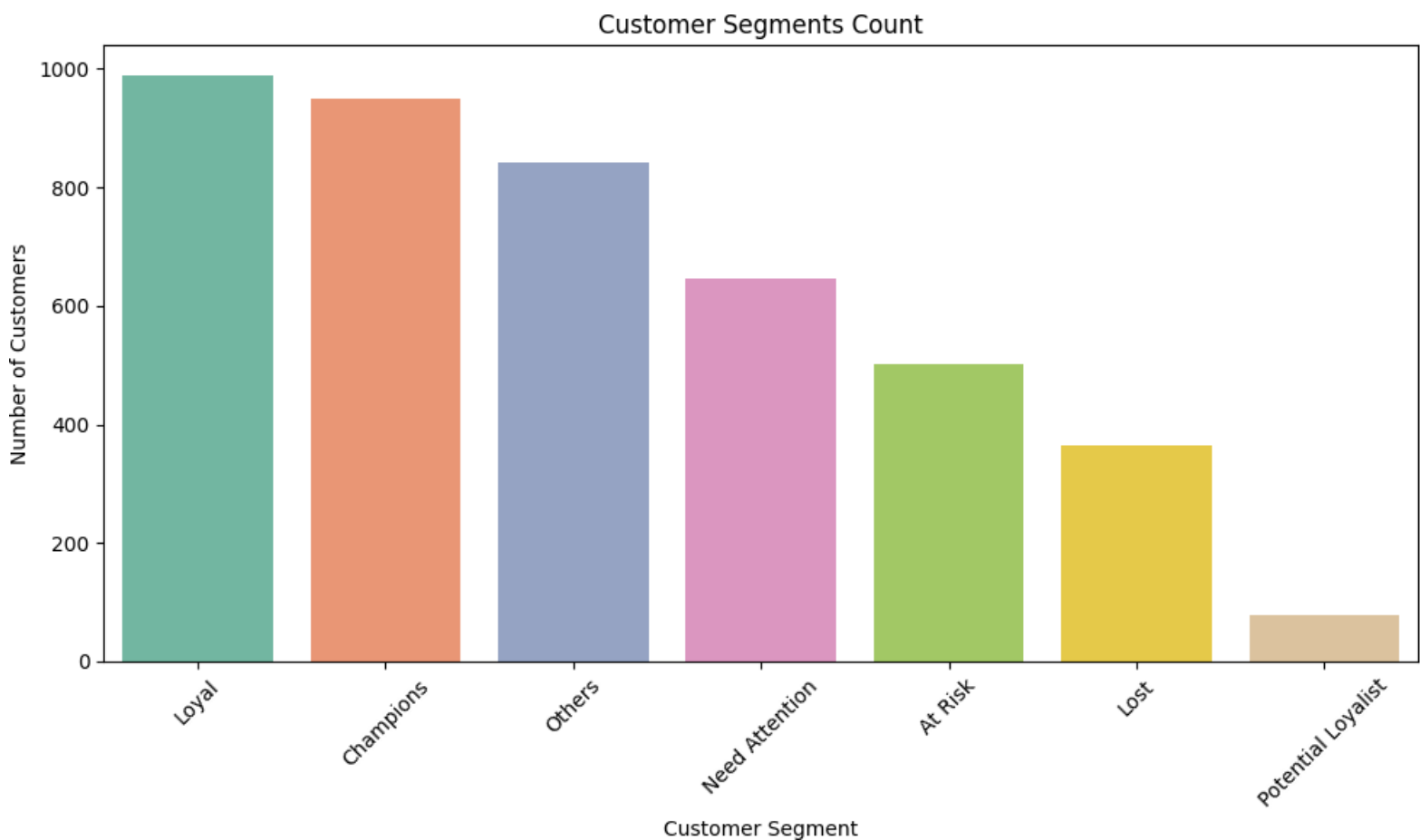
Analysis Results:

The visualization shows that some segments contain a majority of customers, while others are relatively small. This suggests the presence of a few highly valuable segments and many average or low-engagement customers. Businesses can use this insight to target specific segments differently.

Visualization:

Figure: Customer Segment Counts

Bar plot showing the number of customers in each segment, with segments labeled (Champions, Loyal, Potential Loyalist, Need Attention, At Risk, Lost, etc).



Objective : Finding and Validating Clusters Using Elbow Method and Silhouette Score

General Description:

This objective determines the optimal number of customer segments (clusters) using K-Means clustering. Two techniques were used to validate the choice of clusters: the Elbow Method and Silhouette Score (SIH). These help evaluate how well the data points are grouped within clusters.

Specific Requirements:

- Normalize RFM scores using MinMaxScaler to ensure uniform scale.
- Apply K-Means clustering for cluster values ranging from 2 to 10.
- Compute Within-Cluster Sum of Squares (WCSS) for each k-value and plot the Elbow Curve.
- Calculate Silhouette Score for each k-value to assess clustering quality based on cohesion and separation.

Analysis Results:

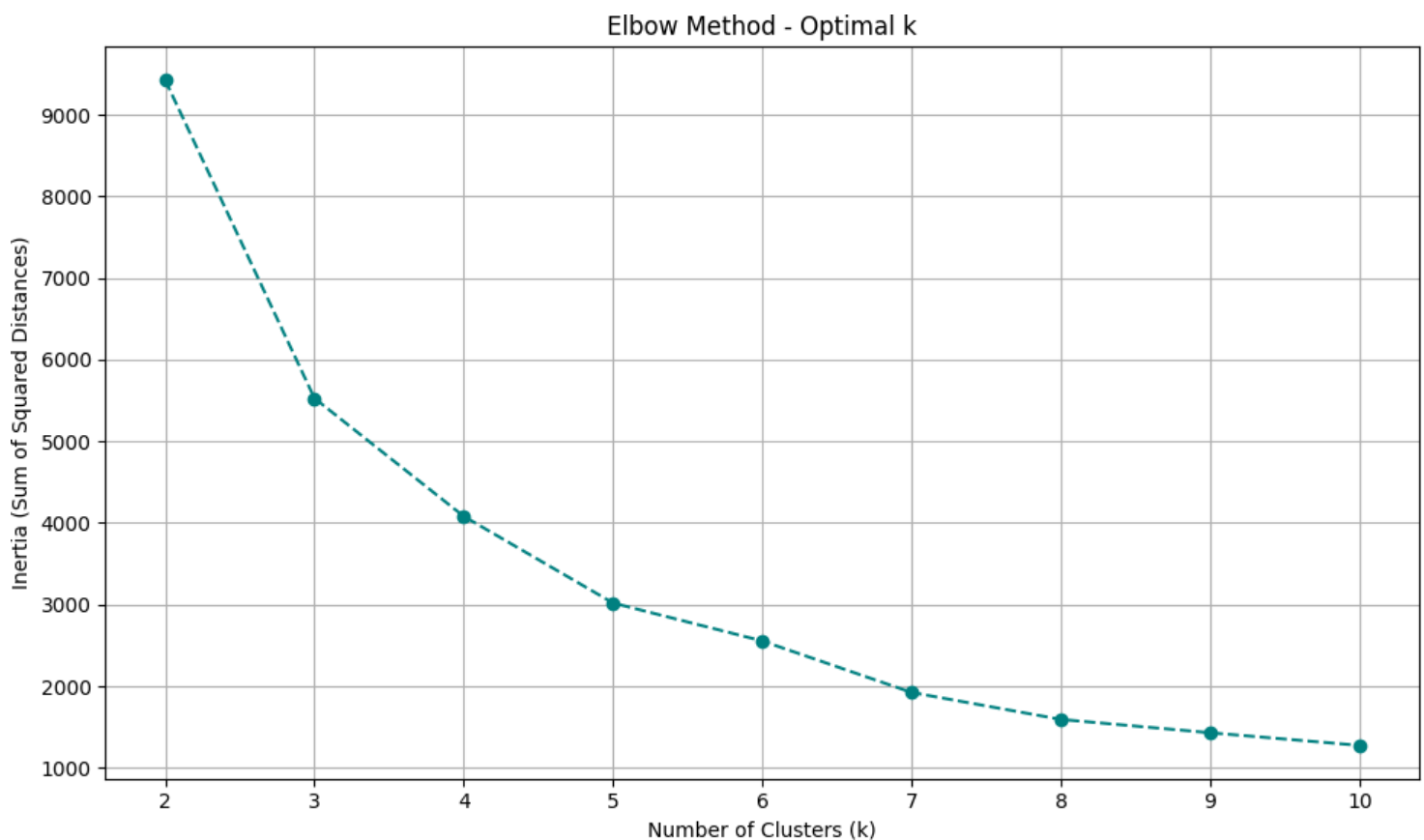
The Elbow Method showed a clear bend around $k = 4$, suggesting diminishing returns for higher cluster counts.

Silhouette Scores peaked at $k = 4$, indicating strong and well-separated clusters at this value.

These results together support the selection of 4 as the optimal number of customer segments.

Visualization:

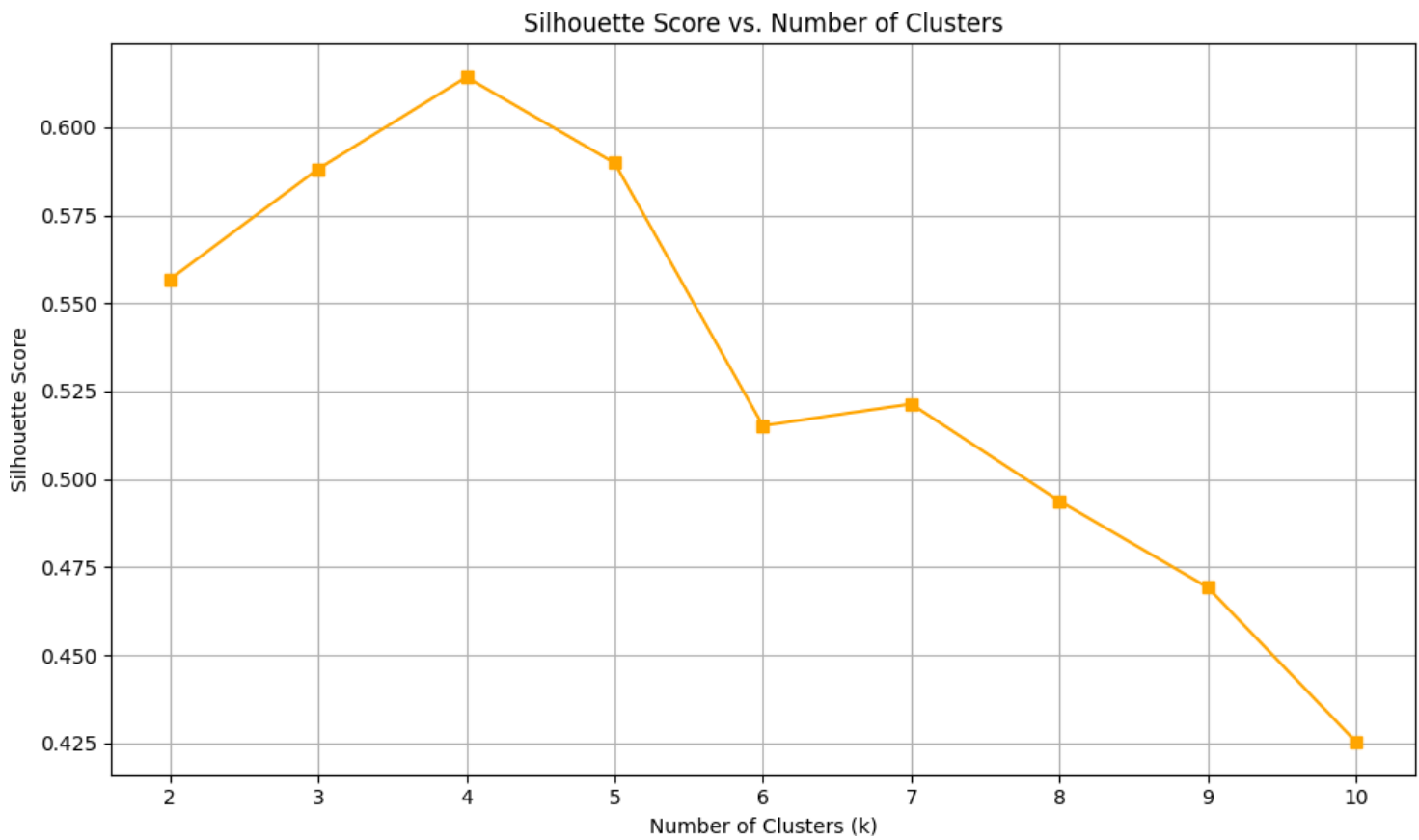
Figure 1: Elbow Plot



- X-axis: Number of Clusters (k)
- Y-axis: WCSS
- Shows a sharp bend at $k = 4$

Figure 2: Silhouette Score Plot

- X-axis: Number of Clusters (k)
- Y-axis: Silhouette Score
- Peaks at $k = 4$, confirming well-formed clusters



Objective : Applying K-Means Clustering for Customer Segmentation

General Description:

After identifying the optimal number of clusters ($k = 4$), this objective focuses on applying K-Means clustering to segment customers based on normalized RFM scores. Each resulting cluster represents a distinct group of customers with similar purchasing behaviors.

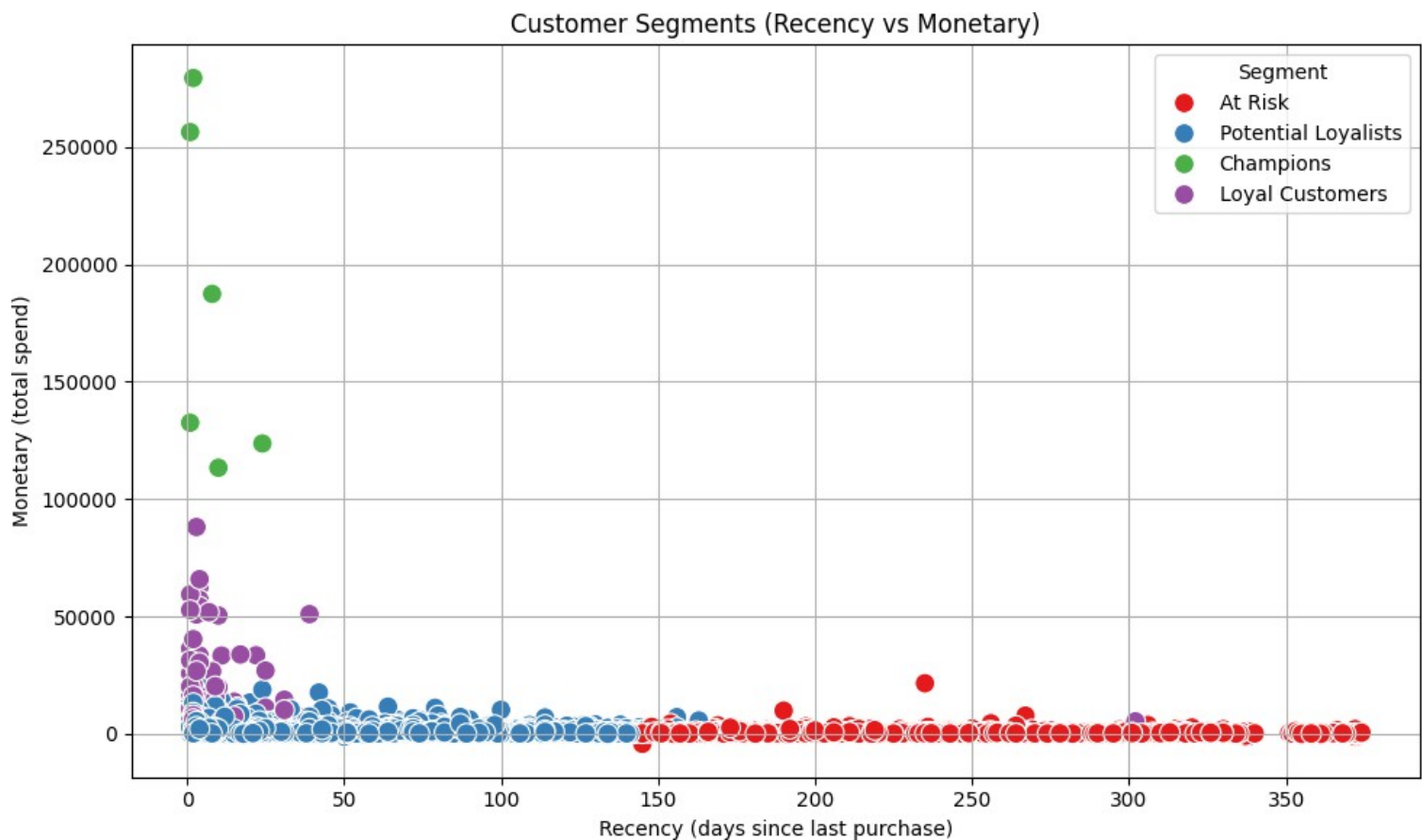
Specific Requirements:

- Use normalized RFM values (Recency, Frequency, Monetary) as input for clustering.
- Apply K-Means with $n_clusters = 4$ based on previous validation.
- Assign each customer to a cluster and append this label to the main dataset.
- Analyze the characteristics (mean RFM values) of each cluster.

Analysis Results:

The customer base was successfully segmented into four clusters:

- **Cluster 0:** Potential Loyalists.
- **Cluster 1:** At Risk.
- **Cluster 2:** Champions.
- **Cluster 3:** Loyal Customers.



These clusters provide a foundation for targeted marketing, retention strategies, and personalized outreach based on actual behavior patterns.

Visualization: Customer Segments Recency vs Monetary

Objective: RFM Metrics Correlation

General Description:

The goal of this objective is to explore the relationships between the key RFM (Recency, Frequency, Monetary) metrics. Understanding how these metrics correlate with each other helps to identify patterns in customer behavior, such as whether frequent buyers tend to spend more or if customers with longer recency are likely to purchase less.

Specific Requirements:

- Calculate the correlation between the three RFM metrics: Recency, Frequency, and Monetary.
- Use a heatmap to visualize the correlation matrix.
- Analyze the strength and direction of the correlations to derive insights.

Analysis Results:

- A positive correlation between Frequency and Monetary suggests that customers who purchase more frequently also tend to spend more.
- A negative correlation between Recency and both Frequency and Monetary indicates that customers who have purchased more recently tend to be more frequent buyers and higher spenders.
- These insights can inform marketing strategies, helping to prioritize customers with high Frequency and Monetary metrics while re-engaging those with high Recency.

Visualization:

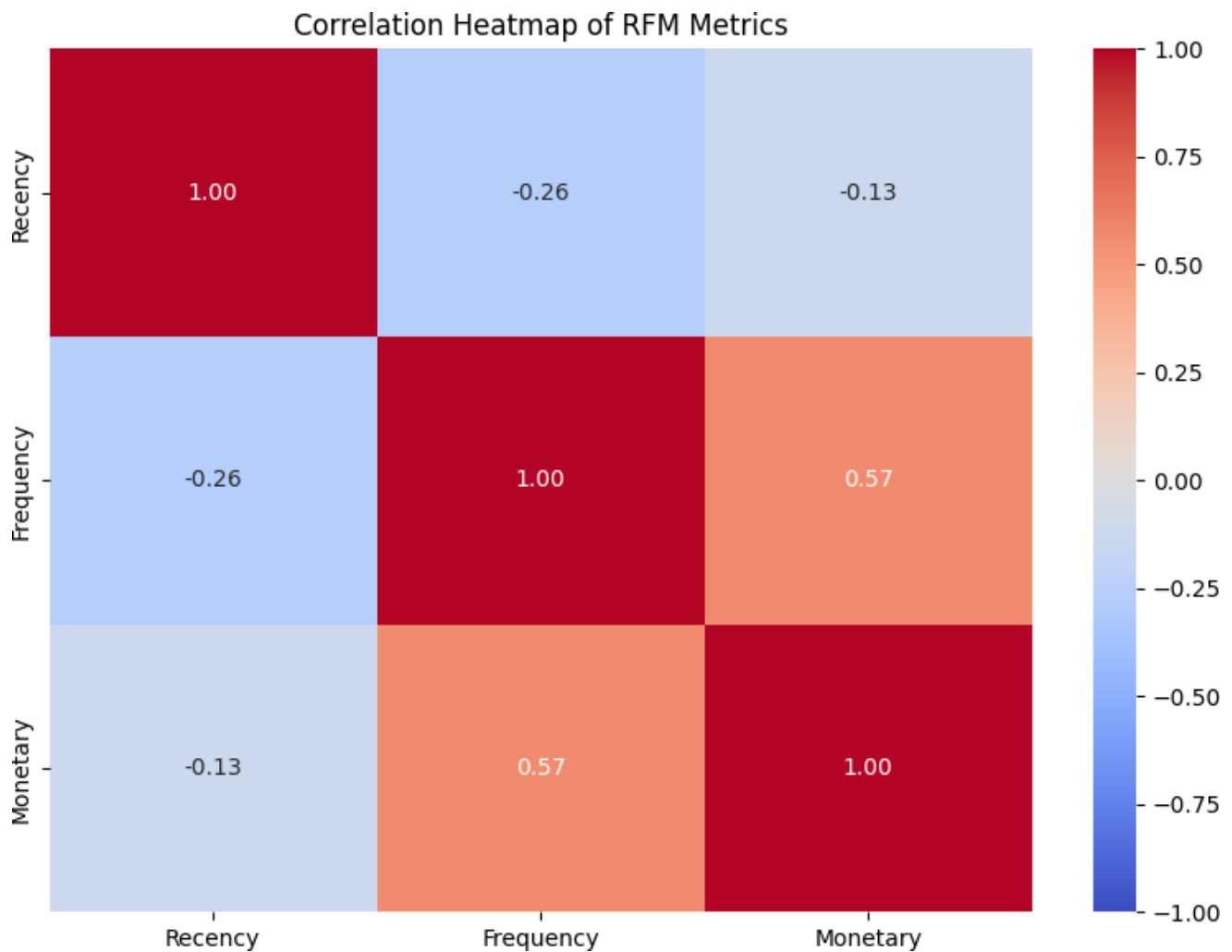
The correlation heatmap visually represents the relationships between the three RFM metrics: Recency, Frequency, and Monetary. The values in the heatmap range from -1 to 1, where:

- A value close to 1 indicates a strong positive correlation, meaning the two metrics move in the same direction (e.g., as one increases, the other does too).
- A value close to -1 indicates a strong negative correlation, meaning the metrics move in opposite directions.
- A value near 0 suggests little to no correlation between the metrics.

In this heatmap:

- The positive correlation between **Frequency** and **Monetary** implies that customers who make frequent purchases tend to spend more.

- The negative correlation between **Recency** and both **Frequency** and **Monetary** suggests that more recent customers are likely to make more purchases and spend more.



Objective : Monthly Purchase Frequency by Customer Segment

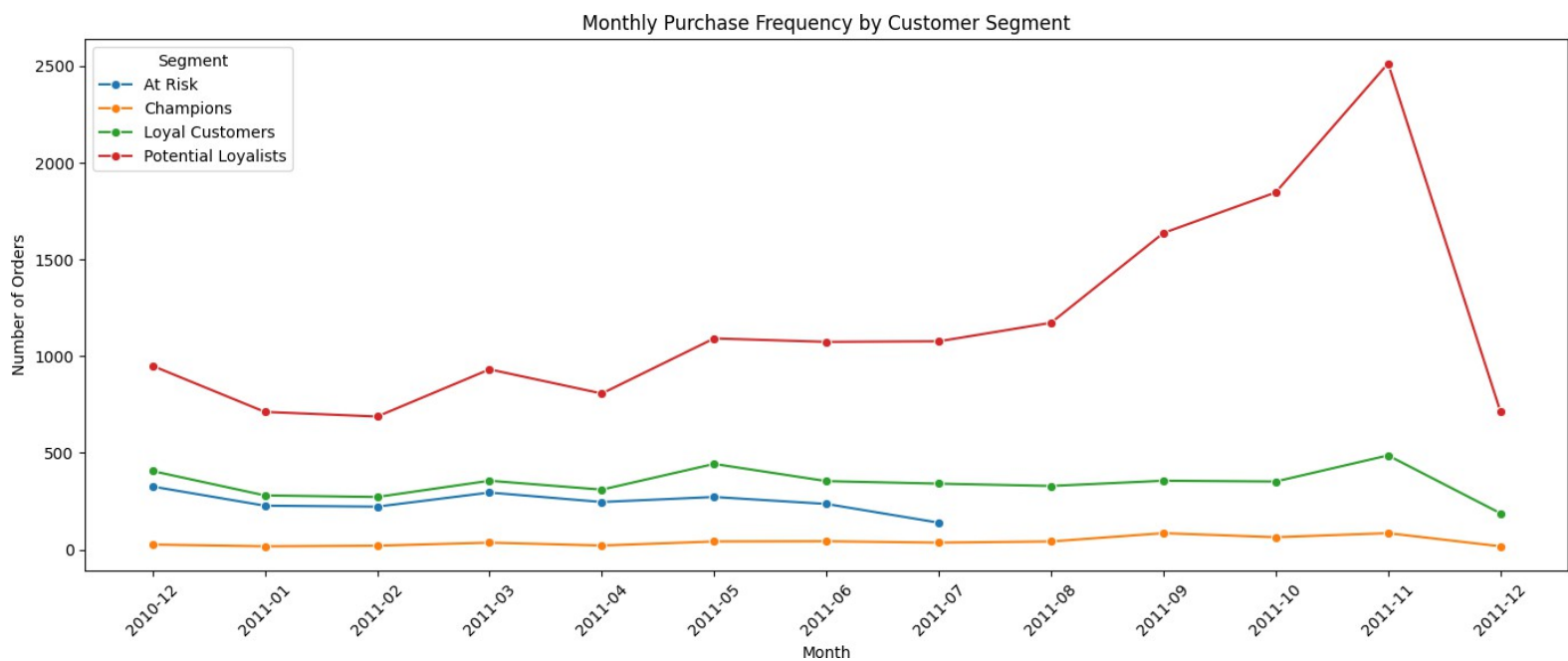
General Description: This objective analyzes the purchase frequency of different customer segments over time by examining the number of unique orders placed each month. Grouping the data by customer segment and month helps to identify trends in how often customers from each segment make purchases.

Specific Requirements:

- Group the data by Month and Segment.
- Count the number of unique InvoiceNo values to capture monthly purchase frequency.
- Create a line plot to visualize the trend of monthly purchase frequency by segment.

Analysis Results: The analysis shows how customer segments, such as "Champions" and "Loyal Customers," exhibit consistent purchase activity over time, while segments like "At Risk" and "Potential Loyalists" show more variability in their purchase behavior.

Visualization: A line plot depicting the monthly purchase frequency for each customer segment, with color-coded segments to highlight changes in behavior over time.



Objective: Visualizing Customer Segment Distribution

General Description:

This objective aims to visualize the distribution of customers across different segments based on their RFM scores. Understanding how customers are spread across these segments helps in formulating targeted marketing and retention strategies.

Specific Requirements:

Customers were segmented into various groups based on their RFM scores. A pie chart was created to show the proportion of customers in each segment, allowing for a visual understanding of the distribution.

Analysis Results:

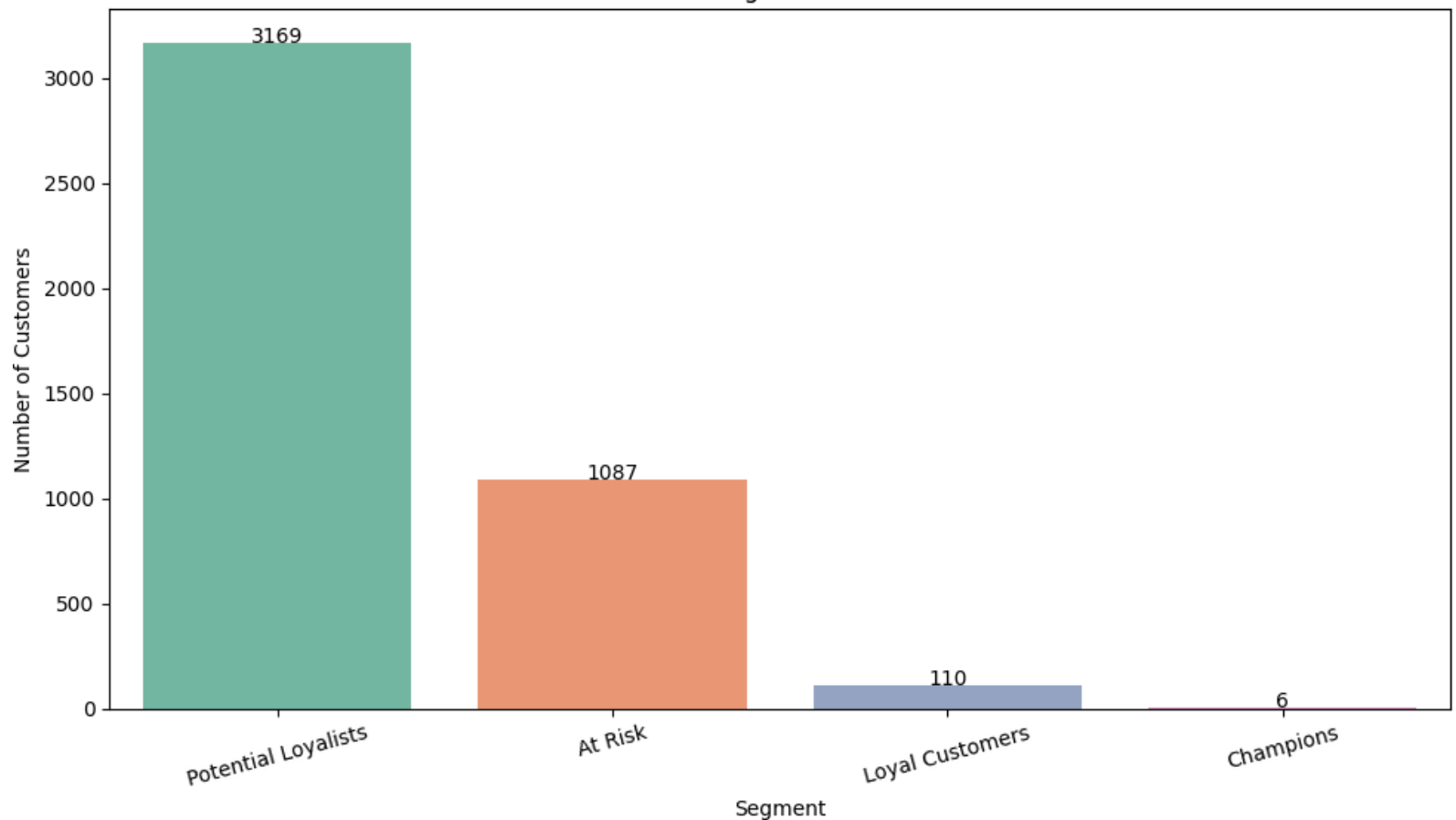
The visualization indicates the percentage of customers in each segment, with some segments being more populated (e.g., Potential Loyalists, At Risk) and others being smaller (e.g., Champions, Loyal). This distribution can help identify which segments are the most valuable and which ones may need more attention.

Visualization:

Figure: Customer Segment Distribution

Pie chart showing the proportion of customers in each segment (Champions, Loyal, Potential Loyalist, At Risk, Lost).

Customer Segment Distribution



Objective: Correlation Heatmap of RFM Features

General Description:

This objective examines the relationships between the three RFM metrics—Recency, Frequency, and Monetary—to understand how these factors interact in customer behavior.

Specific Requirements:

The correlation matrix was computed using Pearson correlation. A heatmap was generated to visualize the strength and direction of relationships among Recency, Frequency, and Monetary.

Analysis Results:

The heatmap shows that Frequency and Monetary have a strong positive correlation, suggesting that customers who purchase more frequently tend to spend more. Recency has a negative correlation with both Frequency and Monetary, indicating that recent buyers are also likely to be frequent and high-spending.

Visualization:

Figure: RFM Feature Correlation

Heatmap showing the correlation coefficients between Recency, Frequency, and Monetary.



Conclusion

This [project](#) aimed to uncover valuable customer insights by performing RFM (Recency, Frequency, Monetary) analysis and applying K-Means clustering to segment customers based on their purchasing behavior. By analyzing transactional data, the study revealed clear patterns in customer engagement and spending, helping to differentiate high-value customers from low-activity ones.

The segmentation process involved calculating total spending and order frequency per customer, identifying top spenders, and understanding customer behavior over time. RFM scores were computed and used to

classify customers into meaningful segments such as Champions, Loyal Customers, Potential Loyalists, At Risk, and Lost. These segments were validated using the Elbow Method and Silhouette Score to ensure appropriate clustering.

The analysis found that a small percentage of customers, especially those in the Champions and Loyal segments, account for a large portion of the company's revenue. Conversely, segments like At Risk and Lost showed signs of disengagement and low contribution, signaling opportunities for reactivation or retargeting. Monthly and seasonal trends also revealed customer purchase behaviors over time, aiding in campaign planning and inventory strategies.

Visualizations such as distribution plots, cluster scatter plots, and segment-wise monthly trends enhanced the interpretability of the analysis, providing clear evidence of how customer behavior differs across segments.

By leveraging this segmentation, businesses can implement focused marketing strategies—for instance, rewarding Champions to maintain loyalty, re-engaging At Risk customers with incentives, or nurturing Potential Loyalists into long-term buyers. This targeted approach supports more efficient resource allocation, improves customer satisfaction, and ultimately drives business growth.

The project demonstrates the effectiveness of RFM-based segmentation as a tool for customer understanding and strategic decision-making, laying the foundation for future personalization, retention campaigns, and long-term value generation.

Future Scope

Future Scope

The current analysis establishes a strong foundation for understanding customer behavior using RFM-based segmentation and clustering. However, there are several directions in which this project can be expanded and refined for greater impact:

- 1. Incorporating Additional Features:** Beyond RFM metrics, incorporating behavioral data such as browsing history, product categories purchased, time between purchases, and discount sensitivity can enable more nuanced segmentation.
- 2. Predictive Modeling:** Integrating machine learning models to predict customer churn, future purchase probability, or lifetime value can enhance strategic planning. These models can use RFM scores and other behavioral data as features.
- 3. Real-Time Segmentation:** Adapting the segmentation pipeline for real-time or near real-time use allows businesses to respond to customer behavior dynamically, improving personalization and responsiveness.
- 4. Integration with Campaign Performance:** Linking segment-wise marketing campaigns with customer responses and purchase outcomes can validate the effectiveness of targeted efforts and inform future strategies.
- 5. Time-Series Forecasting:** Applying forecasting techniques on segment-level revenue and order patterns can help with demand planning and inventory management, especially during seasonal peaks.

6. **A/B Testing for Engagement Strategies:** Future work can involve deploying different marketing approaches for each segment and comparing their performance using controlled A/B testing, helping optimize retention and acquisition strategies.
7. **Geographical and Demographic Enrichment:** Combining RFM segmentation with geographic or demographic data could reveal location-specific or age-specific patterns, enabling hyper-targeted marketing efforts.
8. **Customer Journey Analysis:** Tracking how customers move across segments over time (e.g., from Potential Loyalist to Champion or from Loyal to At Risk) can help in identifying triggers of both loyalty and disengagement.
9. **Automation and Dashboarding:** Automating the segmentation process and integrating it into an interactive dashboard (e.g., using Power BI or Tableau) can make the insights accessible to non-technical teams for ongoing decision-making.

References

1. Pandas Documentation – Essential library for data manipulation and analysis, used extensively in this project for cleaning, grouping, and calculating RFM metrics for customer segmentation. <https://pandas.pydata.org/docs/>
2. NumPy Documentation – Core library for performing numerical operations, essential for data transformation and scaling in the segmentation process. <https://numpy.org/doc/>
3. Matplotlib & Seaborn Visualization Libraries – Libraries used for generating visualizations such as bar plots, scatter plots, and heatmaps to represent customer segments and patterns in the data. <https://matplotlib.org/> <https://seaborn.pydata.org/>
4. scikit-learn Documentation – Machine learning library used for implementing K-Means clustering to segment customers based on their RFM scores. <https://scikit-learn.org/stable/modules/clustering.html#k-means>
5. Python Documentation – Comprehensive documentation for Python programming language, utilized throughout the project for implementing the necessary functions and features. <https://docs.python.org/3/>
6. Jupyter Documentation – Provides support for the Jupyter notebook environment, which was used to run and document the entire analysis and modeling process. <https://jupyter.org/documentation>
7. IPython Documentation – Interactive computing library used for enhancing the workflow within Jupyter notebooks, particularly for quick testing and debugging of code. <https://ipython.readthedocs.io/en/stable/>